

Unit-3 - P&C.

* **Sum Rule** - If an event can occur in 'm' ways and another event can occur in 'n' ways, & if these events cannot occur simultaneously, then one of the two events can occur in $m+n$ ways.

* **Product Rule** - If an event 'h' is decomposed into 2 different tasks then, if an event 'a' can occur in 'm' ways & event 'b' can occur in 'n' different ways then the task 'h' can be performed in $m \times n$ ways.

Q. A bookshelf has '6' different English books, '8' diff French books & '10' different German books. Find in how many ways the full selection can be made.

- i) 3 books from each language
- ii) One book in any one of the language

Ans i) $6 \times 8 \times 10$
 ii) $6 + 8 + 10 = 24$ ways

Q. Suppose a person has 3 shirts & 5 ties, how many diff. types ways of choosing shirt & a tie is possible.
 $\Rightarrow 3 \times 5 = 15$ ways

Q. Suppose we wish to construct a seq of four symbols in which first two are English letters, next two are single digit nos. If no letter or digit ~~can~~ be repeated. Find the no. of diff seq which can be constructed.

$\Rightarrow$

26	25	10	9
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 $\Rightarrow$
letters digit

Q. Suppose a restaurant sells 6 South Indian dishes; 4 North Indian dishes, 3 hot dishes & 2 cold dishes for. If a student wishes to buy a South Indian dish and a hot dish or one North Indian and a cold dish; how many choices will she have?

$$\Rightarrow SI \times H + NI \times C.$$

$$6 \times 3 + 4 \times 2 = 26$$

Q. There are 20 married couple in a party. Find the number of ways in choosing one woman and one man from the party such that they are not married.

$$\Rightarrow 20 \text{ couples} \begin{cases} \rightarrow 20W \\ \rightarrow 20M \end{cases} \Rightarrow 20 \times 19 = 380$$

Q. A license plate consists of 2 English letters which has only 'A', 'E', 'I', 'O', 'U' followed by 4 digits. If repetitions are allowed, then how many license plates will have A-E-I-O-U & even digits.

Ans $\underbrace{\boxed{5} \boxed{5}}_{\text{letters}} \underbrace{\boxed{5} \boxed{5} \boxed{5} \boxed{5}}_{\text{digits}} = 5^6 \text{ plates}$

Q. Find the no of possible arrangements of the letters of the full words.

i) DATABASES -

ii) PEPPER -

iii) DATABASES = $\frac{9!}{3! \cdot 2!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4^2}{2}$

$$\frac{4 \times 2 \times 7 \times 2}{2}$$

$$\frac{52 \times 7 \times 2}{4}$$

Q. i) Find no of permutations of the word MASSASAUGA.
 An = how many of these all 4 A's are together.
 ii) How many of them begin with 'S'

Ans i) $\frac{10!}{3! \times 4!} \Rightarrow \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5}{6} = 25200$

$$\begin{array}{r} 72 \times 35 \\ 360 \\ \hline 2160 \\ \hline 25200 \end{array}$$

ii) $\frac{AAAA}{3!} \times 6! \Rightarrow \frac{7!}{3!} \Rightarrow 840$ $\begin{matrix} M \rightarrow 1 \\ A \rightarrow 4 \\ S \rightarrow 2 \end{matrix}$

iii) ~~$\frac{10!}{4! \times 3! \times 2! \times 1!}$~~ $\Rightarrow \frac{9!}{2! \times 2!}$ $\begin{matrix} U \rightarrow 1 \\ S \rightarrow 1 \end{matrix}$

Q. In how many ways can 6 men & 6 women be seated in a row

- i) If any person may sit next to any other.
 ii) If men & women must occupy alternately.

WM WM WM WM WM WM
 MW MW MW MW MW MW

Ans i) $12!$
 ii) $(6! \times 6!) \times 2 = 720 \times 720 \times 2 = 1036800 \times 2 = 2073600$

Q. 4 diff math books, 5 different CS books, 2 different English books are to be arranged in a shelf.

How many arrangements are possible :-

- i) If books in each particular subj must be together.
 ii) Only math books are together.

Ans i) $\frac{14!}{5! \times 4! \times 2!} \Rightarrow 3! \times 5! \times 4! \times 2!$
 $= 6 \times 120 \times 24 \times 2 = 1440 \times 24 = 34560$

$$\begin{array}{r} 144 \times 24 \\ 3456 \end{array}$$

$M \rightarrow M_1, M_2, M_3, M_4$
 $C \rightarrow C_1, C_2, C_3, C_4, C_5$
 $E \rightarrow E_1, E_2$

$$\begin{aligned} \text{ii) } & \left(\frac{4!}{5! \times 2!} \times 7! \right) \Rightarrow \frac{720 \times 7 \times 8}{5 \times 2 \times 2} = \frac{720 \times 28}{20} \\ & \Rightarrow 24 \times (8!) = 24 \times 720 \times 56 \\ & = \underline{9,67,680} \end{aligned}$$

Q. An airhostess is ~~having~~ ^{hosting} a dinner party for new members of her committee. Because of the size of her home; she can invite only 11 out of 20. How many possibilities are there for her.

Ans. $\Rightarrow {}^{20}C_{11} = \frac{20!}{11! \times 9!} = \underline{167960}$

Q. A gym trainer must select must select 9 girls from the seniors & juniors. If there are 25 seniors & 28 juniors. How many selection can be done.

i) If 2 juniors & 1 senior must be in a team,
 iii) Team must have 4 juniors & 5 seniors.

Ans. i) ${}^{53}C_9$

iii) $\Rightarrow {}^{28}C_2 \times {}^{25}C_1 \times {}^{250}C_6$

ii) $\Rightarrow {}^{28}C_4 \times {}^{25}C_5$

1/2/24 Q. A gym trainer must make 4 volleyball teams of 9 boys from 36 boys. In how many ways can he select these 4 teams

Ans. ${}^{36}C_9 \times {}^{27}C_9 \times {}^{18}C_9 \times {}^9C_9$

Q. Find out all the arrangements of the word TALLAHASSEE & how many should not have will not have adjacent

Ans. All arrangements = $\frac{11!}{3! \times 2! \times 2! \times 2!} = \frac{180}{360} \times 7 \times 8 \times 9 \times 10 \times 11$
 $\frac{8 \times 2 \times 2 \times 2}{2 \times 1 \times 1 \times 1}$

$\Rightarrow 19600 - 9000 = 831600$

A's together = $[3] \times 8! = \frac{9!}{2! \times 2! \times 2!} =$

Alt. $\uparrow T \uparrow L \uparrow L \uparrow H \uparrow S \uparrow S \uparrow E \uparrow E \uparrow$

$\Rightarrow \frac{8!}{2! \times 2! \times 2!} \times {}^9C_3$

The number of ways to distribute n indistinguishable objects into r distinct groups is given by $\boxed{{}^{n+r-1}C_{r-1}}$

* The no. of combinations of n objects taken r at a time with repetition is given by:
 $C(n+r-1, r) \Leftrightarrow {}^{n+r-1}C_{r-1}$

Q. A donut shop offers 20 kinds of donuts. Assuming that there are atleast a dozen of each kind, when we enter the shop, in how many ways can he select a dozen donuts.

Ans. $n=20; r=12$

$\therefore$ Required ans = ${}^{31}C_{12} = 14,11,20,525$

Q. In how many ways we distribute 7 bananas & 6 oranges among 4 children so that each child receives atleast one banana

Ans $7B \rightarrow 4 \Rightarrow 3B.$

New $n=4$ (children); $r=3$ (remaining B).
 $\Rightarrow n$

$$\frac{6C_3 \times 9C_3}{}$$

$$\rightarrow 2! - 12$$

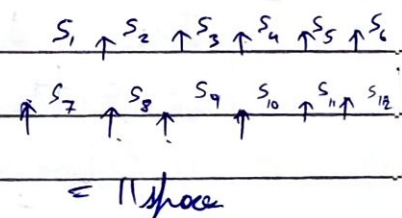
$$45 \quad B_1 B_2 \dots B_{45}$$

Q. A msg is made up of 12 different symbols & is to be transmitted thro a common channel. In addition to 12 symbols the transmitter will also send 45 blank spaces b/n the symbols w/ atleast 3 spaces b/n each pair of consecutive symbols. In how many ways can a transmitter send such msg.

Ans $r=12$; $n=45$; ~~9~~

No of ~~space~~ positions for spaces $= r-1 = 11.$

No of min spaces $= 33$ (11×3 atleast)



$$\therefore \text{Left out spaces} = 45 - 33 = 12.$$

• Arrangement of symbols $= 12!$

$\therefore r=12$; ~~atleast~~ $n=11 \Rightarrow 12$ spaces to be inserted into 11 spaces

$$\Rightarrow \text{total possible ways} = {}^{22}C_{12}$$

$$\therefore \text{Total possible msgs} = 12! \times {}^{22}C_{12}$$

$$\Rightarrow \frac{12! \times 22!}{10! \times 12!} = \frac{22!}{10!} =$$

Q. ~~10 marble~~ In how many ways can 10 marbles be distributed among 5 students.

- i) w/o no restriction
- ii) Each student gets atleast 1 marble

Ans i) ~~$n = 10$ (m)~~ ; ~~$r = 5$ (students)~~
 $n = 10$ (marbles) ; $r = 5$ (students) ; $x = 10$ (marble — item being distributed)
 $\Rightarrow {}^{n+r-1}C_r = {}^{14}C_5 = 1001$

ii) Each student gets atleast 1 marble $\Rightarrow 5$.

Q. President Helen has 4 VPs. She wishes to distribute among them \$1000 in a Christmas bonus check, where each will be written for a multiple of \$100.

i) Allowing the situation where one or more VPs get nothing, President is making selection of size 10 from a collection of size 4 w/ repetition. In how many of size 4 can the president distributions

ii) If there are no hard feeling, each VP should receive. Now, the President is forced with making a selection of size 6 with some collection of size 4. Calculate the no of ways the P

iii) If each VP must get atleast \$100 & and 1 VP or the executive VP gets \$500. In how many ways can the president distribute the check

Ans i) $n = 4$ (VPs) ; $r = 10$ (cheques) $\Rightarrow {}^{13}C_4 = 286$

ii) $n=4$ (VPs) here $r=10-4=6$.

$$\Rightarrow {}^{n+r-1}C_r \Rightarrow {}^9C_6$$

iii) $n=4$ (VPs) here $r=10-5-3=2$.

$$\Rightarrow {}^{n+r-1}C_r = {}^5C_2$$

5/2/24 $\rightarrow$ Binomial Theorem:-

If x & y are variables & n is a +ve integer, then
 $(x+y)^n = {}^nC_0 x^0 y^n + {}^nC_1 x^1 y^{n-1} + {}^nC_2 x^2 y^{n-2} + \dots + {}^nC_n x^n y^0$.

$$\therefore (x+y)^n = \sum_{k=0}^n {}^nC_k x^k y^{n-k}$$

Q. Find the coefficient of $x^5 y^2$ in the expansion of $(x+y)^7$.

Ans

$$\Rightarrow {}^7C_5 \Rightarrow \frac{7 \times 6}{2} = 21$$

$$T_{n+1} = {}^nC_n x^n y^{n-n}$$

$$\text{Stepwise:- } (x+y)^n = \sum_{k=0}^n {}^nC_k x^k y^{n-k}$$

Let $k=n-r$.

$$\text{Given } n=7. \Rightarrow (x+y)^7 = {}^7C_{7-r} x^{7-r} y^{7-(7-r)}$$

$$\Rightarrow \text{Now } X \cdot x^5 y^2 = {}^7C_{7-r} \cdot x^{7-r} y^{7-(7-r)}$$

$$5=7-r \Rightarrow r=2$$

$$\therefore X = {}^7C_5 =$$

Q Find coefficient of $a^5 b^2$ in the expansion of $(2a-3b)^7$.

$$T_{n+1} = {}^nC_n x^n y^{n-n}$$

Ans Let $x=2a$ & $y=-3b$.

$$\text{Now } (x+y)^7 = \sum {}^7C_k x^k y^{7-k}$$

Let $k = n - r \Rightarrow {}^{7-r}C_{7-r} x^{7-r} y^{7-(7-r)}$

$\Rightarrow {}^{7-r}C_{7-r} (2a)^{7-r} (-3b)^{7-(7-r)}$

We need coefficient for $a^5 b^2$.

$\Rightarrow 7-r=5 \Rightarrow r=2$

$\Rightarrow {}^7C_5 (2a)^5 (-3b)^2 \Rightarrow \frac{7 \times 6}{2} \times 32 \times 9 \times a^5 b^2$
 $= 6048 a^5 b^2$

Q Find the coefficient of $x^{12} y^{13}$ in $(x+y)^{25}$.
 Ans $(x+y)^{25} = {}^{25-r}C_{25-r} x^{25-r} y^{25-(25-r)}$

Now $25-r=12 \Rightarrow r=13$

$\Rightarrow {}^{25}C_{12} x^{12} y^{13} \Rightarrow \text{Ans} = {}^{25}C_{12} = 5200300$

→ Multinomial Theorem :-

For the positive integers n, k the coefficients of $x_1^{n_1} x_2^{n_2} \dots x_k^{n_k}$ in the expansion of $(x_1 + x_2 + x_3 + \dots + x_k)^n$ is

$\frac{n!}{n_1! n_2! n_3! \dots n_k!}$ where each n_i is an integer with $0 \leq n_i \leq n$ and $n_1 + n_2 + n_3 + \dots + n_k = n$.

Q Find the coefficients of $(x^2, y^2, z^3), (xyz^5), (x^3 z^4)$ in the expansion $(x+y+z)^7$

Ans For $x^2 y^2 z^3$; $n_1=2, n_2=2, n_3=3$; $n=7$

$\Rightarrow \text{coefficient} = \frac{7!}{2! \cdot 2! \cdot 3!} = 210$

② Multinomial Theorem:

Statement of the Multinomial Theorem:

For any positive integer n and any $k \geq 2$,

$$(x_1 + x_2 + \cdots + x_k)^n = \sum_{a_1 + a_2 + \cdots + a_k = n} \frac{n!}{a_1! a_2! \cdots a_k!} x_1^{a_1} x_2^{a_2} \cdots x_k^{a_k},$$

where:

- The summation is over all non-negative integer solutions $(a_1, a_2, \dots, a_k)$ to the equation $a_1 + a_2 + \cdots + a_k = n$.
- $a_1, a_2, \dots, a_k$ are the exponents of $x_1, x_2, \dots, x_k$, respectively.
- $\frac{n!}{a_1! a_2! \cdots a_k!}$ is the **multinomial coefficient**, which counts the number of ways to distribute n items into k groups of sizes $a_1, a_2, \dots, a_k$.

• For xyz^5 ; coefficient = $\frac{7!}{1!1!5!} \Rightarrow \underline{\underline{42}}$

• For x^3yz^4 ; coefficient = $\frac{7!}{3! \cdot 1! \cdot 4!} \Rightarrow \frac{7 \times 6 \times 5}{6} = \underline{\underline{35}}$

Q. Find the term containing $x^n \cdot y^n$ in the expansion $(2x^3 - 3xy^2 + z^2)^6$

Ans. $\frac{6!}{n_1! n_2! n_3!} \Rightarrow (2x^3)^{n_1} \cdot (-3xy^2)^{n_2} \cdot (z^2)^{n_3}$

• We need to find out the coefficients for $x^n \cdot y^n$.

$\Rightarrow n_3 = 0$

$\Rightarrow 2^{n_1} \cdot (-3)^{n_2} \cdot (x)^{(3n_1+n_2)} \cdot y^{2n_2} \cdot (z^2)^0$

By comparing;

$\Rightarrow 3n_1 + n_2 = 11 \quad \& \quad 2n_2 = 4$

$\Rightarrow n_2 = 2$

$3n_1 + 2 = 11 \Rightarrow \underline{\underline{n_1 = 3}}$

$\Rightarrow \text{Coefficient} = \frac{6!}{3! 2! 0!} \times 2^{n_1} \times (-3)^{n_2} = \frac{6 \times 5 \times 4}{2} = \underline{\underline{60}} \times 2^3 \times (-3)^2$

$\Rightarrow 60 \times 8 \times 9$

$= 60 \times 72 = \boxed{4320}$

Q. Find coeff of xyz^2 in $(2x - y - z)^4$.

Ans. $\frac{4!}{n_1! n_2! n_3!} \Rightarrow (2x)^{n_1} \cdot (-y)^{n_2} \cdot (-z)^{n_3}$

We require, xyz^2 $\therefore n_1 = 1; n_2 = 1; n_3 = 2$

$\Rightarrow \text{Coefficient} = \frac{4!}{1! \cdot 1! \cdot 2!} \times 2 \times (-1)^1 \times (-1)^2 \Rightarrow 12 \times 2 \times -1 \times 1 = \underline{\underline{-24}}$

$a^2 b^3 c^2 d^5 (a + 2b - 3c + 2d)^{16}$

Q. Find coeff of $a^2 b^3 c^2 d^5$ in $(a+2b-3c+2d+5)^{16}$

Ans $\frac{16!}{n_1! \cdot n_2! \cdot n_3! \cdot n_4! \cdot n_5!}$; now $16 = n_1 + n_2 + n_3 + n_4 + n_5$
 $\Rightarrow n_5 = 16 - (2+3+2+5) = 4$

$$\text{Coefficient} = \frac{16!}{2! \cdot 3! \cdot 2! \cdot 5! \cdot 4!} \times (1)^2 \cdot (2)^3 \cdot (-3)^2 \cdot (2)^5 \cdot (5)^4$$

$$\Rightarrow \frac{16!}{2! \cdot 3! \cdot 2! \cdot 5! \cdot 4!} \times (1 \times 8 \times 9 \times 32 \times 625)$$

$$\Rightarrow 4.3589 \times 10^{14}$$

→ Rook Polynomial:-

Consider a board that represents a full chess board or a part of it. Let 'n' be the number of squares present in the board. Two rooks placed on a board are said to be in non capturing positions if they are not in same row or same column for $2 \leq k \leq n$.

Let r_k denote the no of ways in which 'k' rooks can be placed on a board such that no two rook capture each other.

The polynomial $1 + r_1 x + r_2 x^2 + r_3 x^3 + \dots + r_n x^n$ is called the rook polynomial for the board considered.
 • If the board is denoted by 'C'; then the polynomial is represented as $r(C, x)$.
 • $r_0 = 1$ always

Q. Find the rook polynomial for the given board below.

➤ Rook polynomial for $n \times n$ board is given by

$$r(C_{n \times n}, x) = 1 + \binom{n}{1} x + 2! \times \binom{n}{2} x^2 + 3! \times \binom{n}{3} x^3 + \dots + n! \times \binom{n}{n} x^n$$

Expansion formula: In a given board C , suppose we choose a particular square and mark it as $(*)$. Let D denote the board obtained from C by deleting the row and column containing the square $(*)$ and E be the board obtained from C by deleting only the square $(*)$. Then the rook polynomial for the board C is given by $r(C, x) = x r(D, x) + r(E, x)$. This is known as Expansion formula for $r(C, x)$.

Product formula: Suppose a board is made up of two parts C_1 and C_2 where C_1 and C_2 have no squares in the same row or in the same column of C . Such parts are called disjoint sub boards of C . Then the rook polynomial for the board C is given by $r(C, x) = r(C_1, x) \times r(C_2, x)$

This is known as product formula for $r(C, x)$.

Arrangements with forbidden positions: Suppose ' m ' objects are to be arranged in ' n ' places where $n \geq m$. Suppose there are constraints under which some objects cannot occupy certain places. Such places are called the forbidden positions for the said objects. The number of ways of carrying out this task is given by,

$$\bar{N} = S_0 - S_1 + S_2 - S_3 \pm \dots + (-1)^n S_n$$

where $S_0 = n!$, $S_k = (n-k)! \times r_k$ for $k = 1, 2, 3, \dots, n$.

r_k is the coefficient of x^k in the rook polynomial of the board of ' m ' rows and ' n ' columns whose squares represent the forbidden places.

→ $n!$ = number of squares in the board.

→ Normally if $m \geq n$ n becomes greater than number of rows then $r_n = 0$

3	2	1
4	X	X
X	5	6

3	2	1
4		
	5	6

3	2	1
4		
	5	6

Ans. • X, shaded or blank $\Rightarrow$ Rook cannot be placed there.

• $r_0 = 1$ (always)

• $r_1 = 6 \rightarrow$ total no of elements in the board excluding r_1 .

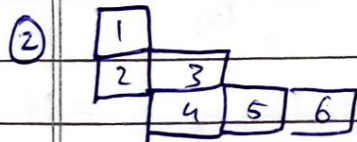
• $r_1 = \{1, 2, 3, 4, 5, 6\}$

• $r_2 = \{(1,4), (1,5), (2,4), (2,6), (3,5), (3,6), (4,5), (4,6)\} \rightarrow 8$.

• $r_3 = 2 \quad (\{1,4,5\}, \{2,4,6\})$

• $r_4 = 0$

$\therefore r(c, x) = 1 + 6x + 8x^2 + 2x^3$



Ans. $r_0 = 1$; $r_1 = 6$

$r_2 = \{\{1,3\}, \{1,4\}, \{1,5\}, \{1,6\}, \{2,4\}, \{2,5\}, \{2,6\}, \{3,5\}, \{3,6\}\}$

$\Rightarrow a$

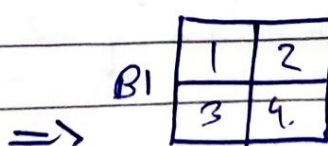
$r_3 = \{\{1,3,5\}, \{1,3,6\}\}$

$r_4 = 0$

$\therefore r(c, x) = 1 + 6x + 9x^2 + 2x^3$

7/2/2024 ③

1	2	X	X	X
3	4	X	X	X
X	X	X	5	6
X	X	X	7	8
X	X	X	9	10 11



(Decompose into smaller boards)

$$B1: r_0 = 1; r_1 = 4; r_2 = 2 \left(\{1,4\}, \{2,3\} \right); r_3 = 0$$

$$r(B1, x) = 1 + 4x + 2x^2$$

$$B2: r_0 = 1; r_1 = 7; r_2 = \left(\{5,8\}, \{5,9\}, \{5,11\}, \{6,7\}, \{6,9\}, \{6,10\}, \{7,9\}, \{7,11\}, \{8,9\}, \{8,10\} \right) = 10$$

$$r_3 = \left(\{5,8,9\}, \{6,7,9\} \right) = 2; r_4 = 0$$

$$r(B2, x) = 1 + 7x + 10x^2 + 2x^3$$

$$(1 + 4x + 2x^2)(1 + 7x + 10x^2 + 2x^3) = (1 + 8x + 7x^2 + 10x^3 + 2x^4) + (4x + 28x^2 + 40x^3 + 8x^4) + (2x^2 + 14x^3 + 20x^4 + 4x^5)$$

$$\Rightarrow 1 + 11x + 40x^2 + 56x^3 + 28x^4 + 4x^5$$

→ Principle of Inclusion & Exclusion

Consider a set S with $|S|$ (cardinality of S) = N . & conditions C_i in which $1 \leq i \leq t$, each of which may be satisfied by some of the elements of S .

The no. of elements of S that satisfy none of the condition C_i is denoted by $\bar{N} = N(\bar{C}_1, \bar{C}_2, \dots, \bar{C}_t)$

$$= S_0 - S_1 + S_2 - S_3 + \dots - S_t$$

where $S_0 = N$

S_1 = summation of each of the condi.

S_2 = summation of 2 condi put together

S_3 = " " 3 condi put together

; & likewise.

Q. Determine the no. of the integers 'n' where $1 \leq n \leq 100$

Q) Principle of inclusion and exclusion:

PRINCIPLE OF INCLUSION AND EXCLUSION:

The addition principle or the principle of inclusion exclusion for 2 sets is

$$\text{given by } |\overline{A \cup B}| = |S| - (|A| + |B|) + |A \cap B|$$

where S is a finite set.

The principle of inclusion-exclusion for 'n' sets is given by

$$|\overline{A_1 \cup A_2 \cup \dots \cup A_n}| = |S| - \sum |A_i| + \sum |A_i \cap A_j| - \sum |A_i \cap A_j \cap A_k| \pm \dots + (-1)^n |A_1 \cap A_2 \cap \dots \cap A_n|$$

$\begin{matrix} \downarrow & \downarrow & \downarrow \\ 2 & 3 & 4 \end{matrix}$

$$|\overline{A_1 \cup A_2 \cup A_3}| = 50 - 51 + 22 - 3$$

$$n \times \begin{matrix} 2, 1, 5 \\ 1 \quad 1 \quad 1 \\ A \quad B \quad C \end{matrix} \quad 1 \leq n \leq 100$$

$$n(A) = 50, \quad n(B) = 37, \quad n(C) = 20$$

$$\therefore 100 - (50 + 37 + 20) + (11 + 11 + 11) - 3$$

$$\begin{aligned} n(A \cap B) &= 11 \\ n(B \cap C) &= 6 \\ n(A \cap C) &= 11 \end{aligned}$$

$$n(A \cap B \cap C) = 3$$

& n is not divisible by 2, 3 or 5.

Ans $C_1 = \div \text{by } 2; C_2 = \div \text{by } 3; C_3 = \div \text{by } 5.$

~~$N(C_1) = N(C_2) = N(C_3) = S_0 = 100.$~~

} S_0

~~$N(C_1) = N(C_2) = N(C_3)$~~

$N(\overline{C_1} \cdot \overline{C_2} \cdot \overline{C_3}) = N(S) - [N(C_1) + N(C_2) + N(C_3)]$

$\cdot N(\overline{C_1}) = 50$

$\cdot N(C_1) = 33$

; $N(C_2) = 20$

} S_1

~~$N(\overline{C_1} \cdot \overline{C_2} \cdot \overline{C_3}) = 100 - 100 = 0$~~ $N(C_1, C_2) = 100/6 = 16.$

$N(C_1/C_2) = 100/10 = 10$

$N(C_2/C_3) = 100/15 = 6.$

} S_2

$N(C_1 C_2 C_3) = 100/30 = 3$ } S_3

$\therefore N(\overline{C_1} \cdot \overline{C_2} \cdot \overline{C_3}) = 100 - (50 + 20 + 33) + (16 + 10 + 6) - (3)$

$\Rightarrow 100 - 103 + 32 - 3 = 26$

8/8/2024

Q. Let S refer set of 100 students in an engineering program; at a college. C_1 : 35 stu at the college among 100 stu are enrolled for course 1. C_2 : 30 stu $\rightarrow$ course 2. If 9 of these stu are enrolled in both course 1 & course 2, find the no of stu who haven't registered for any of the courses.

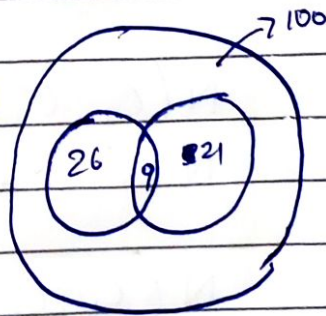
Ans

$N(\overline{C_1} \cdot \overline{C_2}) = S_0 - S_1 + S_2$

$= N - [N(C_1) + N(C_2)] + N(C_1 \cdot C_2)$

$\Rightarrow 100 - [35 + 30] + 9$

$= 100 - 65 + 9 = 44$



* Additional part of Q:-

$C_3 = 30 \text{ stu} \rightarrow$ course 3. If 11 students have registered

for both course 1 & course 3 & 10 stu & for course 2 & course 4 & 5 stu for all the 3 courses. Find how many have registered for any

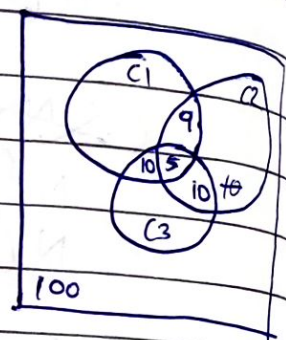
$$N(\bar{C}_1, \bar{C}_2, \bar{C}_3) = N S_0 - S_1 + S_2 - S_3$$

$$\Rightarrow N - [N(C_1) + N(C_2) + N(C_3)] + [N(C_1, C_2) + N(C_2, C_3) + N(C_1, C_3)] - [N(C_1, C_2, C_3)]$$

$$\rightarrow 100 - [35 + 30 + 30] + [9 + 11 + 10] - 5$$

$$= 100 - 95 + 30 - 5$$

$$\Rightarrow 5 + 25 = \underline{\underline{30}}$$



X. Q. In how many ways can 26 letters alphabets be permuted so that none of the patterns:- starts with:
 CAR, DOG, PUN, or BYTE occurs

Ans

$$N = 26!$$

$$N(C_1) = C_1 : \text{CAR} ; N(C_1) : 24! \Rightarrow (26-3) + 1$$

$$N(C_2) : 24!$$

$$N(C_3) : 24!$$

$$N(C_4) : 23!$$

Starts with CAR, DOG, PUN, BYTE

$$N(C_1, C_2) = \frac{(26-6)+2}{2} = 22!$$

$$N(C_1, C_3) = 22! ; N(C_1, C_4) = 21!$$

$$N(C_2, C_3) = 22! ; N(C_2, C_4) = 21!$$

$$N(C_3, C_4) = 21! ;$$

$$N(C_1, C_2, C_3) = (26-9) + 3 = 20!$$

$$N(C_1, C_3, C_4) = 19!$$

$$N(C_1, C_2, C_4) = 19!$$

$$N(C_2, C_3, C_4) = 19!$$

$$N(C_1, C_2, C_3, C_4) = [(26-13)+4]! \Rightarrow 17!$$

$$\begin{aligned} N(\bar{C}_1, \bar{C}_2, \bar{C}_3, \bar{C}_4) &= S_0 - S_1 + S_2 - S_3 \\ &= 26! - (24! + 24! + 23!) + (22! + 22! + 21!) - 22! + 22! + 21! \\ &\quad - (20! + 19! + 19! + 19!) + 17! \end{aligned}$$

$$= (4.0329) - (\cancel{0.01266}) + (0.00002299) - (\dots) \times 10^{26}$$

$$= \underline{4.0202 \times 10^{26}}$$

Q. Find the no of P from A to Z from which none of the patterns ~~from~~ SPIN, GAME, PATH, NET occurs.

Ans $N = 26!$

$$N(C_1) = 23! ; N(C_2) = 23! ; N(C_3) = 23! ; N(C_4) = 24!$$

$$N(C_1, C_2) = (26-8)+2 \Rightarrow 20!$$

$$N(C_1, C_3) = \cancel{26} - 0$$

< If repetition $\Rightarrow 0$ >

$$N(C_1, C_4) = 0$$

$$N(C_2, C_3) = 0$$

$$N(C_2, C_4) = 0$$

$$N(C_3, C_4) = 0$$

$$N(C_1, C_2, C_3) = 0$$

$$N(C_1, C_2, C_4) = 0$$

$$N(C_1, C_3, C_4) = 0$$

$$N(C_2, C_3, C_4) = 0$$

$$N(C_1, C_2, C_3, C_4) = 0.$$

$$N(C_2, C_3, C_4) =$$

$$\Rightarrow N(\bar{C}_1, \bar{C}_2, \bar{C}_3, \bar{C}_4) = S_0 - S_1 + S_2 - S_3$$

$$= 26! - (23! + 23! + 23! + 24!) + 20! - 0.$$

$$= \underline{4.0259 \times 10^{26}}$$

Q. Determine the no of permutations of the pattern JNU IS GREAT such that none of the words "JNU", "IS", "GREAT" occur as consecutive letters.

Ans $N = 10!$; $C_1: \text{JNU}$; $C_2: \text{IS}$; $C_3: \text{GREAT}$.
 $N(C_1) = 8!$; $N(C_2) = 9!$; $N(C_3) = 6!$
 $N(C_1, C_2) = 6!$; $N(C_1, C_3) = 3!$; $N(C_2, C_3) = 4!$
 $N(C_1, C_2, C_3) = 1$

$$\begin{aligned} N(\bar{C}_1, \bar{C}_2, \bar{C}_3) &= S_0 - S_1 + S_2 - S_3 \\ &= 10! - (8! + 9! + 6!) + (6! + 3! + 4!) - 1 \\ &= \underline{\underline{32,25,629}} \end{aligned}$$

2/2/24 → Generalization Principle:-

Consider a set w/ cardinality $|S| = N$ with conditions $C_1, C_2, \dots, C_t$ satisfied by some of the elements of S , the inclusion & exclusion provides a way to determine $N(\bar{C}_1, \bar{C}_2, \dots, \bar{C}_t)$ i.e., no of elements in S that satisfy none of the t conditions in order to determine E_m which denotes no of elements in S , that satisfy exactly m of t conditions.

$$E_1 = N(\bar{C}_1, \bar{C}_2, \bar{C}_3, \dots, \bar{C}_t) + N(\bar{C}_1, \bar{C}_2, \dots, \bar{C}_{t-1}) + N(\bar{C}_1, \bar{C}_2, C_3, \dots, \bar{C}_t) + \dots + N(\bar{C}_1, \bar{C}_2, \bar{C}_3, \dots, C_t)$$

$$E_2 = N(C_1, \bar{C}_2, \bar{C}_3, \dots, \bar{C}_t) + N(\bar{C}_1, C_2, \bar{C}_3, \dots, \bar{C}_t) + \dots + N(\bar{C}_1, \bar{C}_2, \dots, C_{t-1}, C_t)$$

• Keywords used are 'exactly' & 'atleast'.

Note: 1) The above expression can be written as

$$\left| \overline{A_1 \cup A_2 \cup \dots \cup A_n} \right| = S_0 - S_1 + S_2 - S_3 \pm \dots + (-1)^n S_n$$

where $S_0 = |S|, S_1 = \sum |A_i|, S_2 = \sum |A_i \cap A_j|, S_3 = \sum |A_i \cap A_j \cap A_k|, \dots, S_n = |A_1 \cap A_2 \cap \dots \cap A_n|$

2) The number of elements in S that satisfy exactly ' m ' of the given ' n ' conditions is given by

$$E_m = S_m - {}^{m+1}C_1 S_{m+1} + {}^{m+2}C_2 S_{m+2} \mp \dots + (-1)^{n-m} {}^nC_{n-m} S_n, (0 \leq m \leq n)$$

3) The number of elements in S that satisfy at least ' m ' of the given ' n ' conditions is given by

$$L_m = S_m - {}^mC_{m-1} S_{m+1} + {}^{m+1}C_{m-1} S_{m+2} \mp \dots + (-1)^{n-m} {}^{n-1}C_{m-1} S_n, (1 \leq m \leq n)$$

$$* E_1 = S_1 - 2S_2 + 3S_3$$

$$E_2 = S_2 - 3S_3$$

$$L_m = S_m - {}^m C_1 S_{m+1} + {}^{m+1} C_2 S_{m+2} - \dots + (-1)^{j-m} {}^{j+1} C_{j-m} S_j$$

$$E_m = S_m - {}^{m+1} C_1 S_{m+1} + {}^{m+2} C_2 S_{m+2} - \dots + (-1)^{j-m} {}^{j+1} C_{j-m} S_j$$

Q. In how many ways can the letters in the word "ARRANGEMENT" be arranged so that

- 1. There are exactly 2 pairs of ^{consecutive} identical letters.
- 2. At least 2 pairs of consecutive identical letters.

Ans. Total no. of arrangements - $\frac{11!}{2! \cdot 2! \cdot 2! \cdot 2!} = 24,94,800$

$$N(C_1) [2 \text{ consecutive } A_s] = \frac{10!}{2! \cdot 2! \cdot 2!} = 45,3600$$

$$N(C_2) [2 \text{ consecutive } R_s] = \frac{10!}{2! \cdot 2! \cdot 2!} = 4,53,600$$

$$N(C_3) [2 \text{ consecutive } E_s] = N(C_4) [2 \text{ consecutive } N_s] = 4,53,600$$

$$S_1 = \frac{10!}{2! \cdot 2! \cdot 2!} \times 4 = \frac{10!}{2!} = 18,14,400$$

$$N(C_1, C_2) [AA, RR] = \frac{(11-4+2)!}{2! \cdot 2!} = \frac{9!}{4} = 90720$$

$$N(C_1, C_3) = N(C_1, C_4) = N(C_2, C_3) = N(C_2, C_4) \\ = N(C_3, C_4) = \frac{9!}{4}$$

$$\therefore S_2 = \frac{9!}{4} \times 6 = 5,44,320$$

$$E_2 = S_2 - 3C_1 S_3 + 4C_2 S_4$$

$$N(C_1 C_2 C_3) = \frac{8!}{2!} = \frac{8!}{2}$$

$$N(C_1 C_2 C_4) = N(C_1 C_3 C_4) = N(C_2 C_3 C_4) = \frac{8!}{2}$$

$$\therefore S_3 = \frac{8!}{2} \times 4 = 80640$$

$$\therefore N(C_1 C_2 C_3 C_4) = \frac{(11-8+4)!}{1!} \Rightarrow 7! = 5040$$

$$\therefore S_4 = 7! = 5040$$

$$\therefore E_2 = S_2 - {}^3C_1 S_3 + {}^4C_2 S_4$$

$$= 544320 - 3 \times 80640 + 6 \times 5040$$

$$= 332640$$

$$ii) L_m = S_2 - {}^2C_1 S_3 + {}^3C_1 S_4$$

$$= 544320 - 2 \times 80640 + 3 \times 5040 = 398160$$

Q. In how many ways can one arrange the letters of word "CORRESPONDENTS" so that

- There are exactly 2 pairs of consecutive identical letters
- At least 3 pairs of consecutive identical letters
- There are no pairs of consecutive letters

Ans. Total = $\frac{14!}{2! \cdot 2! \cdot 2! \cdot 2! \cdot 2!}$

$$\cdot N(C_1) = N(C_2) = N(C_3) = N(C_4) = N(C_5) = \frac{(14-2+1)!}{(2!)^4} = \frac{13!}{(2!)^4}$$

$$S_7 = \frac{13!}{(2!)^4} \times 5 =$$

$$\begin{aligned} \cdot N(C_1, C_2) &= N(C_1, C_3) = N(C_1, C_4) = N(C_1, C_5) \\ &= N(C_2, C_3) = N(C_2, C_4) = N(C_2, C_5) \\ &= N(C_3, C_4) = N(C_3, C_5) = N(C_4, C_5) = \frac{(14-4+2)!}{(2!)^3} = \frac{12!}{8} \end{aligned}$$

$$\cdot S_2 = \frac{12!}{8} \times 10$$

$$\begin{aligned} \cdot N(C_1, C_2, C_3) &= N(C_1, C_2, C_4) = N(C_1, C_2, C_5) = N(C_1, C_3, C_4) \\ &= N(C_1, C_3, C_5) = N(C_1, C_4, C_5) = N(C_2, C_3, C_4) \\ &= N(C_2, C_3, C_5) = N(C_2, C_4, C_5) \\ &= N(C_3, C_4, C_5) = \frac{(14-6+3)!}{(2!)^2} = \frac{11!}{4} \end{aligned}$$

$$\cdot S_3 = \frac{11!}{4} \times 10$$

$$\begin{aligned} \cdot N(C_1, C_2, C_3, C_4) &= N(C_1, C_2, C_3, C_5) = N(C_1, C_2, C_4, C_5) \\ &= N(C_1, C_3, C_4, C_5) = N(C_2, C_3, C_4, C_5) = \frac{(14-8+4)!}{2!} = \frac{10!}{2!} \end{aligned}$$

$$\cdot S_4 = \frac{10!}{2} \times 5$$

$$\cdot N(C_1, C_2, C_3, C_4, C_5) = (14-10+5)! = 9!$$

$$E_2 = S_2 - {}^3C_1 S_3 + {}^4C_2 S_4 - {}^5C_3 S_5$$

$$= 350179200$$

$$ii) L_3 = S_3 - {}^3C_2 S_4 + {}^4C_2 S_5 = 74753280$$

$$iii) E_0 = S_0 - S_1 + S_2 - S_3 + S_4 - S_5 = 1286046720$$

→ Derangements:

A permutation of n objects in which none of the objects are in its natural/original place

No. of possible derangements of n objects is denoted by d_n & is given by;

$$d_n = n! \times \sum_{k=0}^n \frac{(-1)^k}{k!}$$

$$= n! \times \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + \frac{(-1)^n}{n!} \right]$$

Q. Find the no. of derangements of the nos 1, 2, 3, 4
 List all derangements

Ans. $n=4$

$$d_n = 4! \times \left[1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \right]$$

$$= 9$$

Derangements -

2	3	4	1	3	1	4	2	4	1	2	3
2	4	1	3	3	4	1	2	4	3	1	2
2	4	3	1	3	4	2	1	4	3	2	1

Q. Find the no. of derangements of the no. 1, 2, 3, 4, 5, 6 where the 1st 3 nos should be 1, 2, 3 in any order. List the derangements

Ans. 1, 2, 3 → d_3

$$d_3 = 3! \times \left[1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} \right] = 2$$

$$\text{Total possibilities} = 2 \times 2 = 4$$

Derangements : 3 1 2 6 4 5
 3 1 2 5 6 4
 2 3 1 6 4 5
 2 3 1 5 6 4

* Note - For values of $n \geq 7$, $d_n = n! \times e^{-1}$

Q. For the integers 1, 2, 3, ..., n there are 11660 derangements where 1, 2, 3, 4, 5 appear in first five positions.
 $n = ?$

Ans $d_5 = 120 \times \left[1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} \right] = 44$

Now $d_5 \times d_{(n-5)} = 11660$

$\Rightarrow d_{(n-5)} = 265$

$\Rightarrow 265 = (n-5)! \times e^{-1}$

$(n-5)! = 265 \times e \Rightarrow (n-5)! = 720.399$

$\therefore (n-5)! = 6!$

$\therefore \underline{\underline{n = 11}}$

Q. There are 8 letters to 8 distinct people to be placed in 8 different addressed envelopes. Find the no. of ways of doing this so that atleast 1 letter reaches to the right person.

Ans $d_8 = 8! \times e^{-1} = 14832.899$ $\rightarrow$ None of them got correct letter

Total possibilities = $8! = 40320$ $\rightarrow$ 8 letters in 8 envelopes

$\therefore$ No way atleast one person receives = Total - None

$= 40320 - 14832.899$

$= 25487.100$

$= 25487$

M T W T F S S
☐ ☐ ☐ ☐ ☐ ☐ ☐

COMPASS

Date :

Q. 7 books are distributed to 7 students for reading.
 In how many ways can the books be collected
 & redistributed. In how many ways can the distribution
 be made such that 2 student will get the same book
 in both the times.

Ans. Answer = $7! \times d7 \Rightarrow 7! \times 7! \times e^{-7}$
 $= 9344726.413 \approx 9344726$